MEMVLN: STREAMING EPISODIC MEMORY FOR VISION-LANGUAGE NAVIGATION

Yan Wang[1*], **Jiangyong Huang**[1,2,*], **Siyuan Huang**[1,†], **Baoxiong Jia**[1,†]
[1] State Key Laboratory of General Artificial Intelligence, BIGAI
[2] Peking University
https://memvln.github.io/
[*] Equal contribution. [†] Corresponding authors.

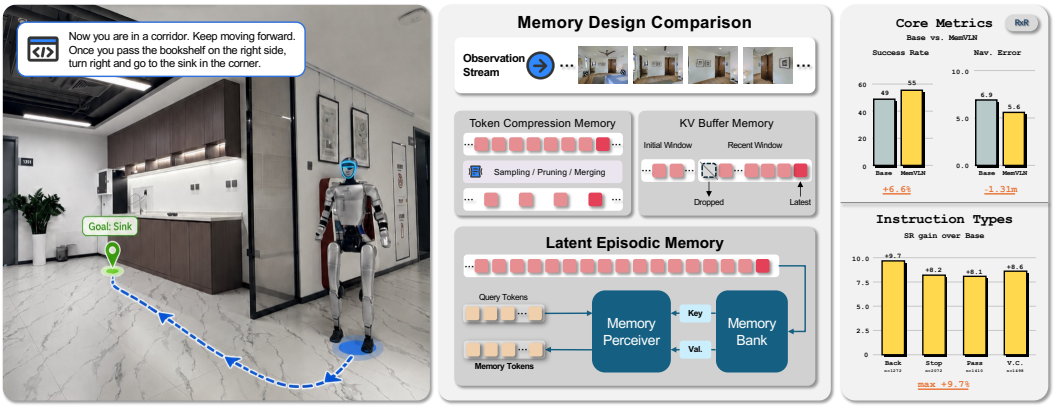


Figure 1: **MEMVLN overview.** We design a streaming episodic memory framework that endows vision-language models (VLMs) with navigation memory for streaming video. MEMVLN employs a memory module that progressively stores compressed perceptual features and produces compact memory tokens via cross-attention. MEMVLN elevates vision-language navigation performance, including long-horizon and real-world scenarios.

ABSTRACT

Vision-language navigation (VLN) requires an embodied agent to follow language instructions through 3D environments. Maintaining persistent episodic memory remains a challenge for VLM-based streaming agents in long-horizon navigation. We present MEMVLN, a streaming episodic memory framework that maintains a bounded bank of compressed perceptual features annotated with timesteps and action-derived relative poses, and produces compact memory tokens via cross-attention. We pair the proposed memory with a frozen pretrained geometry encoder. To learn the memory module, we design a masking-based trajectory summarization task with multi-granularity sub-trajectory decomposition, encouraging route reconstruction exclusively from episodic memory. Under navigation training restricted to the official R2R/RxR trajectories, MEMVLN achieves state-of-the-art performance on most reported metrics on standard VLN-CE benchmarks. Long-horizon analyses further characterize when episodic memory is useful. For physical evaluation, we collect 30 instruction-conditioned episodes in one physical scene, use 24 as demonstrations for post-training, and hold out six for testing. On the held-out episodes, MEMVLN improves success over the VLM baseline from 17% to 40%, and post-training further raises it to 47%.

1 INTRODUCTION

Vision-language navigation (VLN) is a fundamental task in embodied AI, requiring an embodied agent to follow language instructions through 3D environments (Anderson et al., 2018b; Krantz et al.,

2020; Majumdar et al., 2023). Recent vision-language models (VLMs) have substantially improved navigation performance, with monocular streaming video as a prevailing formulation that is natural and practical for real-world deployment (Zhang et al., 2025; Cheng et al., 2025; Wei et al., 2026b;a). Despite rapid progress, the lack of persistent memory remains a critical constraint for real-world long-horizon navigation, where decisions depend on observations from earlier in the episode. In this paper, we address the key problem: *how to design a simple yet effective episodic memory that integrates seamlessly with streaming VLM to elevate VLN capability*?

Navigation memory has received prior attention, yet existing approaches remain insufficient along several key dimensions. Early methods encode navigation history in recurrent hidden states (Elman, 1990; Hochreiter & Schmidhuber, 1997; Hong et al., 2021), compressing perceptual observations into a fixed-size representation. Explicit memory representations retain observation-level features but are often coupled to panoramic observations or topological maps, requiring adaptation to monocular streaming video in continuous environments (Chen et al., 2021b; Mezghan et al., 2022; An et al., 2024; Wang et al., 2023b; Yu et al., 2025). More recent works revisit navigation memory within a streaming formulation (Zeng et al., 2026; Zheng et al., 2025a), yet the capacity limitation persists. These shortcomings motivate our approach: *an episodic memory design that is expressive and natively compatible with streaming video, accompanied by a systematic study of navigation memory spanning long-horizon navigation scenarios and real-world validation*.

We propose MEMVLN, a streaming episodic memory framework that endows VLMs with persistent navigation memory. MEMVLN maintains dense short-term observations for immediate action while building a compact, queryable summary of the accumulated episode history. Concretely, a memory bank progressively stores perceptual features of streaming observations alongside their timestep and action-derived relative pose. For action prediction, a memory perceiver extracts latent features from the bank and encodes them with the corresponding spatio-temporal context. It then derives a fixed-size set of memory tokens via cross-attention with learnable queries grounded in the current timestep and relative pose. Memory tokens enter the VLM's multi-modal sequence for auto-regressive action prediction. When the bank reaches its capacity limit, a frame-merging strategy consolidates consecutive frames into a bounded episodic summary. As a complementary source of spatial features, we use a frozen pretrained StreamVGGT encoder (Zhuo et al., 2026) and train a lightweight adapter to fuse its features into the VLM. Both the memory module and the geometry encoder integrate into existing VLMs for streaming video without modifying the underlying LLM architecture.

To supervise memory learning, we design a dedicated auxiliary task alongside navigation. Specifically, we adopt trajectory summarization (Zheng et al., 2024; Zhang et al., 2024a; Cheng et al., 2025) as the supervision for the memory module, where local visual tokens are masked, leaving memory tokens as the visual evidence for describing the traversed path. To enrich the supervisory signal, we further decompose each full trajectory into hierarchically structured sub-trajectories (Hong et al., 2020), yielding fine-grained supervision across multiple temporal scales. Together, the memory module design, multi-granularity trajectory summarization, and masking strategy constitute a principled framework for learning navigation memory.

We conduct extensive experiments in simulation and on a physical robot. On standard VLN-CE benchmarks (Anderson et al., 2018b; Ku et al., 2020; Krantz et al., 2020), MEMVLN reaches at least 55% success on both R2R and RxR and achieves state-of-the-art performance on most metrics among methods whose navigation fine-tuning trajectories are restricted to the official R2R/RxR training splits. Ablations isolate the contributions of memory, geometry, and auxiliary supervision, while long-horizon analyses further examine when memory is useful. For physical evaluation on the G1 robot, we collect 30 episodes in one physical scene, use 24 as demonstrations for post-training, and hold out six for testing. On the held-out episodes, MEMVLN improves success over the VLM baseline from 17% to 40%, and post-training further raises it to 47%.

2 RELATED WORK

Vision-Language Navigation. VLN requires an embodied agent to follow natural language instructions through 3D environments. Early works establish VLN as a foundational task for embodied AI (Chang et al., 2017; Zhu et al., 2017; Xia et al., 2018; Anderson et al., 2018a; Savva et al., 2019; Anderson et al., 2018b; Chaplot et al., 2020; Qi et al., 2020; Ku et al., 2020; Krantz et al., 2020; Thomason et al., 2020; Ramakrishnan et al., 2021; Krantz et al., 2023; Yadav et al., 2023). Given

scarce annotated navigation data, considerable work targets data-centric approaches, including data augmentation (Tan et al., 2019; Liu et al., 2021; Zhu et al., 2020; Hong et al., 2020; He et al., 2025; Cui et al., 2025) and data scaling in both simulation environments (Hao et al., 2020; Kamath et al., 2023; Wang et al., 2023c) and the real world (Guhur et al., 2021; Linghu et al., 2024; Cheng et al., 2025; Chen et al., 2026). As model capabilities advance, VLN evaluation encompasses increasingly complex scenarios (Majumdar et al., 2024; Zhang et al., 2024b; Yokoyama et al., 2024b; Khanna et al., 2024; Qiao et al., 2025a; Lin et al., 2025; Wang et al., 2025b). In this paper, we focus on the fundamental VLN task in continuous environments and study how navigation memory can be learned and leveraged without relying on massive data scaling (Anderson et al., 2018b; Ku et al., 2020; Krantz et al., 2020; Hong et al., 2020).

Vision-Language Models for Navigation. Effective VLN requires models to perceive 3D environments and comprehend natural language instructions. While early approaches trained navigation models from scratch (Fried et al., 2018; Nguyen et al., 2019; Wang et al., 2019; Hao et al., 2020), the strengths of pretrained VLMs have emerged as a more promising pathway towards general-purpose VLN (Majumdar et al., 2022; Gadre et al., 2023; Dorbala et al., 2022; Zhou et al., 2023; Wang et al., 2024a; Long et al., 2024; Zhou et al., 2024). This paradigm has inspired diverse approaches, including topological structure methods (Chen et al., 2021a; Krantz et al., 2021; Hong et al., 2022; Chen et al., 2022a;b; Yokoyama et al., 2024a; An et al., 2024), explicit 3D spatial modeling (An et al., 2023; Wang et al., 2024b; Huang et al., 2024; Zhu et al., 2025; Wang et al., 2025d; 2026), and trajectory-level modeling (Shah & Levine, 2022; Shah et al., 2023; Sridhar et al., 2024; Wang et al., 2025c; Gode et al., 2025). More recently, the input formulation has been simplified to monocular streaming video, under which navigation performance has improved rapidly (Zheng et al., 2024; Zhang et al., 2024a; 2025; Cheng et al., 2025; Wei et al., 2026b; Zeng et al., 2026; Zheng et al., 2025a; Qi et al., 2025; Wei et al., 2026a; Chu et al., 2026), with significant progress also demonstrated in real-world deployment (Gervet et al., 2023; Yue et al., 2024; Yang et al., 2024; Qiao et al., 2025b; Sun et al., 2026; Yue et al., 2026; Yu et al., 2026; Zhang et al., 2026). Despite this progress, directly retaining long visual histories in streaming VLM policies increases input context and computation, motivating dedicated memory mechanisms.

Memory for Embodied Agents. Memory mechanisms for embodied agents predate modern VLM-based navigation and span several task formulations. Recurrent policies compress navigation history into hidden states (Elman, 1990; Hochreiter & Schmidhuber, 1997; Hong et al., 2021), whereas explicit approaches retain observation histories or structured spatial representations (Chen et al., 2021b; Mezghan et al., 2022; Wang et al., 2023b; Yu et al., 2025). SMT appends each observation to a growing within-episode memory for visual navigation, while E.T. jointly encodes language with the full visual-action history for ALFRED (Fang et al., 2019; Pashevich et al., 2021). ESceme instead maintains a cross-episode, location-indexed scene memory for VLN (Zheng et al., 2025b). Beyond navigation, MemoryVLA retrieves decision-relevant history from a perceptual-cognitive memory for robotic manipulation (Shi et al., 2026). In contrast, MemVLN targets a bounded, episode-local visual bank for streaming VLN, with time/pose-grounded latent readout and memory-specific trajectory-summary supervision.

3 METHOD

Task Formulation. We study vision-language navigation in continuous environments (VLN-CE), where an embodied agent receives streaming observations and follows language instructions through 3D environments. At each timestep t, the agent receives an egocentric observation o_t and a task instruction $\mathcal{T}$, and predicts an action a_t. The objective is to learn a navigation policy $p(a_t \mid o_{1:t}, \mathcal{T})$. Following prior work (Cheng et al., 2025), actions are represented as text comprising an action type $a_t^{\text{type}} \in \{\texttt{move forward}, \texttt{turn left}, \texttt{turn right}, \texttt{stop}\}$ and an action scale a_t^{scale}, where forward distances are in $\{25, 50, 75\}$cm and turn angles are in $\{15°, 30°, 45°\}$.

3.1 MODEL OVERVIEW

As illustrated in Fig. 2, MEMVLN comprises the native vision encoder and large language model (LLM) of Qwen3-VL, a frozen geometry encoder, and a memory module. At navigation step t, the agent acquires a new RGB observation $o_t \in \mathbb{R}^{H \times W \times 3}$, which together with recent observations forms

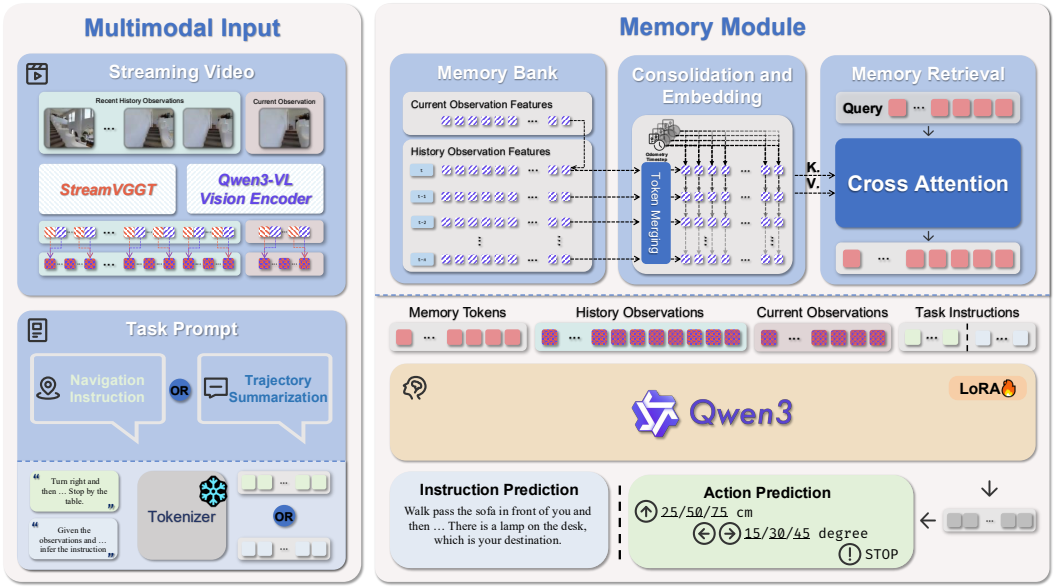


Figure 2: **MEMVLN model overview.** MEMVLN combines Qwen3-VL's native SigLIP-2-based vision encoder and LLM, a frozen StreamVGGT geometry encoder, and our memory module. It receives a short-term context for local perception and an episodic memory for long-horizon reasoning.

the short-term visual context. The native SigLIP-2-based vision encoder π_V encodes this context into visual features, with $f_t \in \mathbb{R}^{H' \times W' \times d}$ denoting the current observation's patch features. We use pretrained visual geometry models to improve the base VLM's spatial understanding. In parallel, frozen StreamVGGT (Wang et al., 2025a; Zhuo et al., 2026) provides camera-token representations for the same context. A lightweight trainable adapter ϕ aligns and fuses the geometry and visual representations, producing local hybrid tokens $\mathcal{H}_t = [h_t^{\text{hist}}, h_t]$ for the recent history and current observation, respectively. Complementing the short-term visual context, the memory module encodes the accumulated visual history into memory tokens m_t, as detailed in Sec. 3.2.

Vision-Language Model Design. MEMVLN augments the vanilla LLM input with memory tokens and geometry features, modeling $p(a_t \mid m_t, \mathcal{H}_t, \mathcal{T})$. To enhance awareness of the current situation, we explicitly distinguish the recent-history tokens h_t^{hist} from the current-observation tokens h_t within $\mathcal{H}_t$. The interleaved multi-modal input to the LLM π_L is:

$$\Big[\{\underbrace{\mathcal{P}_{\text{mem}}}_{\substack{\text{memory}\\\text{prompt}}}, \underbrace{m_t}_{\substack{\text{memory}\\\text{tokens}}}\}, \{\underbrace{\mathcal{P}_{\text{hist}}}_{\substack{\text{history}\\\text{observation}\\\text{prompt}}}, \underbrace{h_t^{\text{hist}}}_{\substack{\text{history}\\\text{observation}\\\text{tokens}}}\}, \{\underbrace{\mathcal{P}_{\text{curr}}}_{\substack{\text{current}\\\text{observation}\\\text{prompt}}}, \underbrace{h_t}_{\substack{\text{current}\\\text{observation}\\\text{tokens}}}\}, \underbrace{\mathcal{T}}_{\substack{\text{task}\\\text{instruction}}}, \underbrace{a_t}_{\substack{\text{action}\\\text{output}}}\Big], \tag{1}$$

$$a_t = \pi_L(m_t, \mathcal{H}_t, \mathcal{T}). \tag{2}$$

The history observation components $\mathcal{P}_{\text{hist}}$ and h_t^{hist} are omitted at the initial step $t = 1$. The dense short-term context anchors fine-grained local perception, while the retrieved memory supplies a bounded summary of accumulated navigation history.

3.2 MEMORY MODULE

Memory Bank. Upon receiving a new observation o_t, the corresponding visual feature f_t is appended to an episode-specific memory bank along with its timestep t and relative pose z_t. The pose encodes position (x, y) and orientation $(\cos\theta, \sin\theta)$ relative to the starting point. At policy inference, it is dead-reckoned solely from the executed discrete-action history; neither ground-truth pose nor external odometry is provided to the policy or memory module. When the bank reaches its maximum capacity, the two consecutive frames with the highest average patch-level cosine similarity are merged, retaining a bounded representative summary. We store the visual feature f_t rather than the hybrid feature h_t.

Memory Perceiver. A memory perceiver resamples the bank into a compact representation using learnable queries and cross-attention (Vaswani et al., 2017; Jaegle et al., 2021). Prior to serving as keys and values, the bank features undergo two operations:

- **Spatial Downsampling.** To reduce computational cost, patch features within each frame are spatially merged using ToMe (Bolya et al., 2023):

$$\bar{f}_{1:t} = \texttt{ToMe}(f_{1:t}). \tag{3}$$

- **Spatio-temporal Embedding.** Timestep and action-derived relative pose are encoded into both the bank features and learnable queries q via sinusoidal temporal embeddings (Vaswani et al., 2017) and Fourier spatial features (Rahimi & Recht, 2007; Tancik et al., 2020; Li et al., 2021):

$$\bar{f}^{+}_{1:t} = \bar{f}_{1:t} + \texttt{sinusoid}(1:t) + \texttt{Fourier}(z_{1:t}), \tag{4}$$

$$q^{+} = q + \texttt{sinusoid}(t) + \texttt{Fourier}(z_t). \tag{5}$$

The memory perceiver then yields a compact and fixed-size set of memory tokens via cross-attention:

$$m_t = \texttt{cross-attn}(q^{+}, \bar{f}^{+}_{1:t}). \tag{6}$$

The memory module offers three complementary properties for VLN: (1) **Long-horizon context**: the bank retains representative visual evidence (*e.g.*, signs, doorways, landmarks) from across the episode, extending the available context beyond recent observations; (2) **Bounded representation**: frame consolidation bounds the number of stored entries, spatial downsampling limits the tokens processed per frame, and the perceiver produces a fixed-size set of memory tokens; and (3) **Spatio-temporal grounding**: timestep and action-derived relative-pose embeddings relate stored evidence to the agent's current point along the trajectory. At inference, the readout queries are instruction-agnostic by design: the memory serves as an episodic record of what the agent has experienced, while the downstream LLM performs instruction-conditioned interpretation.

3.3 Data and Training

MemVLN is trained jointly on two tasks: VLN and trajectory summarization. The navigation task teaches the model to predict the next action, while the summarization task shapes the memory representation towards route-level semantics. Both tasks use a unified cross-entropy loss.

Vision-Language Navigation. VLN trajectories are preprocessed with action merging following NaVILA (Cheng et al., 2025), where consecutive atomic actions within a window of three steps are merged, *e.g.*, three consecutive "`move forward 25cm`" actions become a single "`move forward 75cm`". This enriches action diversity and improves training efficiency. At each step, the model is supervised to predict the ground-truth action a^{*}_{t}:

$$\mathcal{L} = -\log p(a^{*}_{t} \mid m_t, \mathcal{H}_t, \mathcal{T}). \tag{7}$$

Trajectory Summarization. Trajectory summarization provides auxiliary supervision for the memory module. Using the same VLN trajectories, the task instruction $\mathcal{T}$ is repurposed as the ground-truth label. During summarization, all local visual tokens $\mathcal{H}_t$ are masked, leaving the memory tokens m_t as visual evidence. This masking strategy trains the model to reconstruct navigational semantics exclusively from episodic memory, supervising it to retain evidence relevant to long-horizon instruction following, such as landmarks marking completed subgoals or visual cues indicating stopping positions. With the summarization prompt $\mathcal{T}_{(\text{TS})}$, the loss is:

$$\mathcal{L} = -\log p(\mathcal{T} \mid m_t, \mathcal{T}_{(\text{TS})}). \tag{8}$$

Sub-trajectory Derivation. We enrich whole-trajectory summarization with sub-trajectory supervision. FGR2R annotations (Hong et al., 2020) provide sub-instructions and corresponding waypoints in discrete navigation graphs. We map these waypoints to continuous environments by retrieving the nearest points on the continuous trajectory. This yields sub-trajectory splits that provide finer-grained supervision, encouraging the memory module to capture semantics at multiple temporal scales.

Table 1: **Evaluation results on VLN-CE R2R Val-Unseen split.** "Pano.", "Odo.", and "S.RGB" denote panoramic view, odometry, and single RGB; at policy inference, MemVLN instead derives relative pose from executed actions. External Data counts fine-tuning examples beyond those derived from official R2R/RxR trajectories (*e.g.*, EnvDrop (Tan et al., 2019)). Bold marks the best streaming result among methods restricted to those official navigation trajectories. Results are partially referenced from JanusVLN (Zeng et al., 2026).

Method	Observation				R2R Val-Unseen				Training
	Pano.	Odo.	Depth	S.RGB	NE↓	OS↑	SR↑	SPL↑	External Data
CMA (Hong et al., 2022) [CVPR22]	✓	✓	✓		6.20	52.0	41.0	36.0	-
VLN↻BERT (Hong et al., 2021) [CVPR21]	✓	✓	✓		5.74	53.0	44.0	39.0	-
Ego^2-Map (Hong et al., 2023) [ICCV23]	✓	✓	✓		5.54	56.0	47.0	41.0	-
DreamWalker (Wang et al., 2023a) [ICCV23]	✓	✓	✓		5.53	59.0	49.0	44.0	-
AO-Planner (Chen et al., 2025) [AAAI25]	✓		✓		5.55	59.0	47.0	33.0	-
g3D-LF (Wang & Lee, 2025) [CVPR25]		✓	✓	✓	5.70	59.5	47.2	34.6	-
NaVid-4D (Liu et al., 2025) [ICRA25]			✓	✓	5.99	55.7	43.8	37.1	-
NavMorph (Yao et al., 2025) [ICCV25]			✓	✓	5.75	56.9	47.9	33.2	-
NaVid (Zhang et al., 2024a) [RSS24]				✓	5.47	49.1	37.4	35.9	953K
Sim2Real (Wang et al., 2024b) [CoRL24]				✓	5.95	55.8	44.9	30.4	0K
StreamVLN (Wei et al., 2026b) [ICRA26]				✓	6.05	53.8	45.5	41.6	10033K
Uni-NaVid (Zhang et al., 2025) [RSS25]				✓	5.58	53.3	47.0	42.7	3577K
NaVILA (Cheng et al., 2025) [RSS25]				✓	5.37	57.6	49.7	45.5	12574K
JanusVLN (Zeng et al., 2026) [ICLR26]				✓	**5.17**	58.0	52.8	49.2	0K
MemVLN (Ours)				✓	5.19	**61.2**	**55.0**	**50.5**	0K

4 EXPERIMENTS

Implementation Details. We use Qwen3-VL-4B (Bai et al., 2025), including its native SigLIP-2-based vision encoder, together with StreamVGGT-1B (Zhuo et al., 2026). Both encoders are frozen. We adapt the LLM with LoRA (Hu et al., 2022) and train the memory module and geometry adapter from scratch, yielding 1B trainable parameters in total. The recent observation window contains up to 16 frames. The model is trained for 1 epoch (2 days on 8×A100 80G GPUs) using AdamW (Loshchilov & Hutter, 2019) with a base learning rate of 2×10^{-5}, linear warmup, and cosine decay. See detailed hyperparameters in *Appendix*.

Data and Evaluation. We train on the official R2R (Anderson et al., 2018b) and RxR (Ku et al., 2020) training trajectories, comprising 10,819 and 19,996 demonstrations. Full-trajectory summary targets reuse the instructions from these routes; fine-grained R2R supervision uses FGR2R sub-instruction/waypoint annotations derived from R2R routes (Hong et al., 2020). No additional navigation trajectories or external multimodal fine-tuning data are used. For simulation, we roll out the policy in Habitat-Sim (Savva et al., 2019) under the VLN-CE protocol (Krantz et al., 2020) and report Val-Unseen results. For R2R, we report success rate (SR), oracle success rate (OSR), success weighted by path length (SPL), and navigation error (NE). For RxR, we report SR, SPL, normalized dynamic time warping (nDTW) (Ilharco et al., 2019), and NE.

4.1 MAIN SIMULATION RESULTS

Comparison Baselines. We compare primarily against recent VLM-based VLN methods, emphasizing the streaming-video formulation (Wei et al., 2026b; Zeng et al., 2026). For methods with multiple training configurations, Tabs. 1 and 2 report the selected configuration and its additional training-example count. Our state-of-the-art comparison is restricted to methods whose navigation fine-tuning trajectories come only from the official R2R/RxR training splits.

Results. Under this official-trajectory regime, MEMVLN achieves state-of-the-art performance on most metrics in Tabs. 1 and 2. On R2R, MEMVLN reaches 55.0% SR and 50.5% SPL, exceeding JanusVLN by 2.2 and 1.3 percentage points (pp), respectively. OSR rises from 58.0% to 61.2%, with comparable NE (5.19 vs. 5.17 m). On RxR, SR rises from 51.4% to 55.4% (+4.0 pp) and SPL from 44.3% to 48.4% (+4.1 pp). nDTW increases from 59.1 to 64.6, while NE falls from 6.46 to 5.59 m (−0.87 m). These larger RxR gains highlight MEMVLN's effectiveness on longer instructions and more complex trajectories.

Table 2: **Evaluation results on VLN-CE RxR Val-Unseen split.** Settings follow Tab. 1 and JanusVLN (Zeng et al., 2026).

Method	Observation				RxR Val-Unseen				Training
	Pano.	Odo.	Depth	S.RGB	NE↓	SR↑	SPL↑	nDTW↑	External Data
CMA (Hong et al., 2022) [CVPR22]	✓	✓	✓		8.76	26.5	22.1	47.0	-
VLN↻BERT (Hong et al., 2021) [CVPR21]	✓	✓	✓		8.98	27.0	22.6	46.7	-
AO-Planner (Chen et al., 2025) [AAAI25]	✓		✓		7.06	43.3	30.5	50.1	-
NavMorph (Yao et al., 2025) [ICCV25]			✓	✓	8.85	30.8	22.8	44.2	-
Sim2Real (Wang et al., 2024b) [CoRL24]				✓	8.79	36.7	25.5	18.1	0*K*
Uni-NaVid (Zhang et al., 2025) [RSS25]				✓	6.24	48.7	40.9	-	3577*K*
NaViLA (Cheng et al., 2025) [RSS25]				✓	6.77	49.3	44.0	58.8	13132*K*
JanusVLN (Zeng et al., 2026) [ICLR26]				✓	6.46	51.4	44.3	59.1	0*K*
StreamVLN (Wei et al., 2026b) [ICRA26]				✓	6.22	52.9	46.0	61.9	~26330*K*
MemVLN (Ours)				✓	**5.59**	**55.4**	**48.4**	**64.6**	0*K*

Table 3: **Ablations of memory (mem.), geometry (geo.), and auxiliary supervision (aux.).**

Setting	R2R Val-Unseen				RxR Val-Unseen			
	NE↓	OS↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑
Baseline	6.56	51.1	43.9	40.9	6.90	48.8	43.9	59.7
+ mem.	6.38	56.7	45.6	41.6	6.28	53.3	45.9	59.6
+ geo.	6.04	58.1	49.2	45.2	5.87	**56.3**	**49.7**	62.6
+ mem. + geo.	5.85	56.3	49.7	45.8	5.74	56.1	49.4	63.0
+ mem. + geo. + aux.	**5.19**	**61.2**	**55.0**	**50.5**	**5.59**	55.4	48.4	**64.6**

Table 4: **Memory capacity ablations on R2R Val-Unseen: bank size N_b and token count N_m.**

(a) Memory bank capacity

N_b	NE↓	OSR↑	SR↑	SPL↑
0	5.87	57.4	50.9	47.2
100	**5.19**	**61.2**	**55.0**	**50.5**
200	6.24	58.7	50.2	45.7

(b) Memory-token count

N_m	NE↓	OSR↑	SR↑	SPL↑
0	5.87	57.4	50.9	47.2
300	**5.19**	**61.2**	**55.0**	**50.5**
600	6.75	53.9	45.1	40.8

4.2 Ablation Studies

Component Analysis. We ablate memory (mem.), geometry (geo.), and trajectory-summary supervision (aux.) on R2R-CE and RxR-CE in Tab. 3. Memory alone improves success on both benchmarks, raising R2R OSR from 51.1% to 56.7% and RxR SR from 48.8% to 53.3%. Geometry provides complementary spatial information, increasing SR from 45.6% to 49.7% on R2R and from 53.3% to 56.1% on RxR when added to memory. The full model exceeds geometry-only by 5.8 pp SR and 5.3 pp SPL on R2R, while lowering NE from 6.04 to 5.19 m. This highlights the value of learning episodic representations alongside spatial perception.

Trajectory-summary supervision adds 5.3 pp SR and 4.7 pp SPL over memory plus geometry on R2R. On RxR, the full model achieves the lowest NE and highest nDTW among these variants. These R2R gains suggest that summarizing past trajectories helps memory capture navigation behavior useful for subsequent decisions.

Memory Capacity. In Tab. 4, our default setting ($N_b = 100$, $N_m = 300$) improves R2R-CE SR by 4.1 pp and SPL by 3.3 pp over the zero-memory setting, while achieving the best SR and SPL in both capacity ablations. Results on RxR-CE and configuration details are provided in Sec. C.1.

4.3 Long-Horizon Analysis

Settings. We conduct long-horizon analysis on the RxR Val-Unseen split, which features complex instructions and long-horizon navigation episodes. Task complexity is quantified by the number of clauses in each instruction, delimited by "," and ".", and categorized as: simple (1-3), medium (4-6), complex (7-10), and very complex (>10). We evaluate three model variants: the VLM baseline,

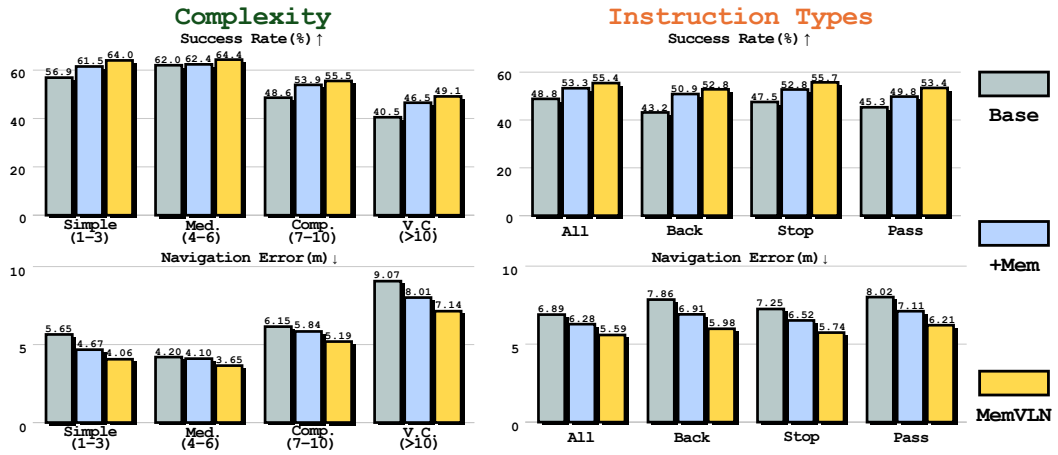


Figure 3: **Long-horizon analysis on RxR-CE Val-Unseen.** Success rate (top) and navigation error (bottom) are grouped by instruction complexity (left) and instruction type (right). Base denotes the VLM baseline, and +Mem adds the memory module. Back, Stop, and Pass denote instructions involving backtracking, stop conditions and long hallways, and passing landmarks, respectively.

baseline with memory module, and the full MEMVLN. Beyond task complexity, we further categorize episodes by whether the instruction involves: (i) backtracking, (ii) explicit stopping conditions and long hallways, and (iii) passing landmarks. We report the success rate (SR) and navigation error (NE) metrics.

Results. In Fig. 3, memory raises SR and lowers NE across all complexity groups and instruction types; the full MEMVLN further improves both metrics. Across complexity levels, memory yields its largest SR gain on very complex instructions: SR rises from 40.5% to 46.5%, while NE falls from 9.07 to 8.01 m. The full model reaches 49.1% SR and 7.14 m NE (+8.6 pp SR over the baseline).

On backtracking episodes, memory raises SR from 43.2% to 50.9% and lowers NE from 7.86 to 6.91 m; the full model reaches 52.8% SR and 5.98 m NE. Memory also gains 5.2 pp and 4.5 pp SR on stop-condition and landmark-passing subsets, respectively. These gains on complex instructions and backtracking highlight the value of episodic context for following extended instructions and revisiting observed locations.

4.4 REAL-WORLD EXPERIMENT

Settings. We deploy on G1 in one physical scene and collect 30 instruction-conditioned episodes with human corrections. We split these episodes into 24 post-training demonstrations and six held-out test episodes. We evaluate the VLM baseline (Base), its memory-augmented variant (+Mem), the full MEMVLN, and the post-trained MEMVLN-PT, executing each held-out instruction five times for 30 trials per model. Post-training uses step-weighted behavior cloning, assigning larger weights to corrected steps. We report success rate (SR) and collision rate (CR), where CR is the fraction of trials involving physical contact, contact-induced blockage, or contact-induced termination.

Results. In Fig. 4, MEMVLN achieves 40.0% SR (12/30 successful trials), up from Base's 16.7% (5/30), while reducing CR from 56.7% to 46.7%. Memory alone raises SR to 30.0% (four additional successes over Base). Compared with +Mem, the full model gains 10.0 pp SR and lowers CR by 20.0 pp. Post-training with the 24 demonstrations further raises SR from 40.0% to 46.7% (14/30 successful trials), while CR remains at 46.7%.

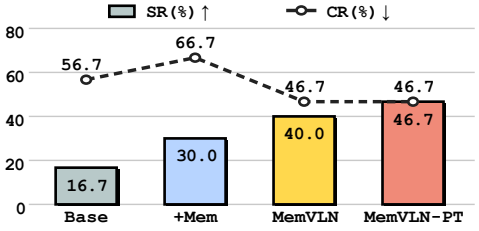


Figure 4: **Real-world results on G1.**

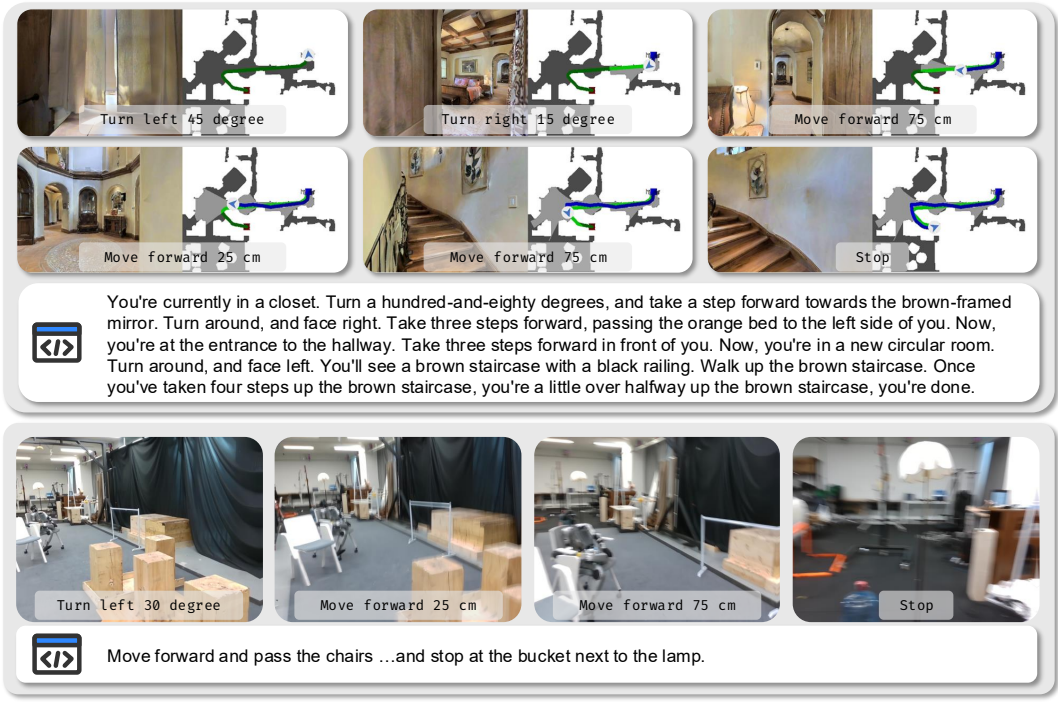


Figure 5: **Qualitative results in simulation and the real world.** Selected observations, actions, and instructions for RxR-CE (top, with top-down trajectories) and G1 (bottom).

Inference Cost. Fixed-input replay on one RTX 4090 yields mean policy latencies of 503.2 ms for Base and 691.0 ms for full MEMVLN. The memory module contributes 300 tokens to the LLM input, independent of route length. Profiling, deployment details, and per-instruction outcomes are provided in Secs. B and C.3.

4.5 QUALITATIVE RESULTS

Figure 5 shows an RxR-CE route from a closet through a hallway to a staircase, and a G1 rollout past chairs, stopping at the bucket beside a lamp. More examples are in Sec. C.4.

5 CONCLUSION

MEMVLN combines plug-and-play episodic memory with a frozen pretrained geometry encoder for streaming vision-language navigation. Multi-granularity trajectory summarization trains the bounded memory representation, masking local visual tokens to reconstruct routes exclusively from episodic memory. With navigation training restricted to official R2R/RxR trajectories, MEMVLN achieves state-of-the-art performance on most reported VLN-CE metrics. Simulation analyses characterize memory's utility across instruction complexities and navigation scenarios. In one physical scene, post-training on 24 demonstrations improves success on six held-out episodes.

6 LIMITATIONS AND FUTURE WORK

First, inference-time retrieval and similarity-based consolidation are instruction-agnostic, so the bounded bank may discard task-relevant observations in long episodes. Second, while trajectory summarization supervises memory learning, broader training strategies remain under-explored (Zheng et al., 2025c). Finally, the real-world study evaluates six held-out episodes in one scene, with substantial collision rates; cross-scene generalization is untested. Future work will investigate task-aware consolidation, broader deployment settings, and safety-aware post-training.

AI USE STATEMENT

During manuscript revision, we used generative AI tools to assist with literature discovery, drafting and editing selected passages based on author-provided materials, discussing experimental design and interpreting results, and developing and debugging research code. These tools also assisted with figure preparation and LaTeX formatting. The authors reviewed and revised the AI-assisted material and take responsibility for the final manuscript and reported results.

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A Additional Implementation Details

This section complements the implementation details in the main paper with the training protocol and memory configuration used for the default MemVLN configuration.

Data Statistics. We summarize the training data statistics in Tab. 5. Each instruction-conditioned demonstration is treated as one episode. For simulation training, the number of navigation samples counts the step-wise action-prediction targets after action merging; the trajectory summarization data is derived from the same episodes. For real-world post-training, we collect 24 demonstrations using an Intel RealSense D455 RGB-D camera mounted on the G1 robot for training. The D455 RGB sensor has a nominal horizontal field of view of 90°. Its RGB frames are captured at 1280×720 and resized to 640×480 before model processing to match the simulation input resolution.

Table 5: **Training data statistics.** Navigation samples count step-wise action-prediction targets after action merging. Simulation sample counts are rounded to the nearest thousand.

Setting	Dataset	Training Data		RGB Input		
		Episodes	Nav. Samples	Resolution	HFOV	Camera
Simulation	R2R Train	10,819	~334K	640×480	90°	Habitat-Sim
	RxR Train	19,996	~1.02M	640×480	90°	Habitat-Sim
Real-world PT	G1 demonstrations	24	1,053	640×480	90°	Intel RealSense D455

Streaming Episode Sampler. Training a stateful memory policy requires temporally coherent data loading. Standard random sampling breaks an episode into unrelated steps, preventing the memory bank from being updated in the same chronological order encountered at inference time. We therefore employ a streaming episode sampler that assigns complete episodes to each device and feeds the steps within each episode sequentially. This protocol allows each device to maintain continuous memory streams while supporting distributed training and task mixing.

Default Memory Configuration. Following the memory module formulation in the main paper, the memory bank stores the visual feature f_t of each observation together with its timestep t and relative pose z_t; at policy inference, z_t is derived from the executed action history. We denote the maximum number of retained frame entries as N_b. Each frame is spatially compressed to at most N_f patch tokens using ToMe (Bolya et al., 2023). A new observation is first appended to the bank. If this increases the bank size to $N_b + 1$, frame merging consolidates the two consecutive entries with the highest average patch-level cosine similarity, reducing the bank size by one and restoring the capacity constraint. The memory perceiver then uses N_m learnable queries to retrieve a fixed-size set of memory tokens m_t. Before each memory readout, the current visual feature is added to the bank; the queries are conditioned only on the current timestep and action-derived relative pose, while the instruction is introduced downstream by the LLM. Since each query produces one output token, N_m is also the number of memory tokens injected into the VLM. The default configuration uses $N_f = 100$, $N_b = 100$, and $N_m = 300$. Additional settings are summarized in Tab. 6.

Simulation Training. As described in the main paper, we adopt Qwen3-VL-4B (Bai et al., 2025) as the VLM backbone and StreamVGGT-1B (Zhuo et al., 2026) as the geometry encoder. The visual encoder and geometry encoder remain frozen. We adapt the LLM with LoRA (Hu et al., 2022), while training the memory module and geometry adapter from scratch. More details are summarized in Tab. 7.

Real-world Post-training. The real-world demonstrations record both the model prediction and the human-corrected executed action. We post-train the policy using the same token-level cross-entropy objective as in simulation, while assigning a larger sample weight to steps requiring correction:

$$\mathcal{L}_{\text{PT}} = \frac{\sum_t w_t \mathcal{L}_t}{\sum_t w_t}, \qquad \mathcal{L}_t = -\log p(a_t^* \mid m_t, \mathcal{H}_t, \mathcal{T}). \tag{9}$$

We set w_t to 3.0 for corrected non-stop actions, 0.5 for actions accepted without correction, and 2.0 for stop actions. This correction-weighted behavior cloning objective focuses adaptation on states where the deployed policy requires human intervention.

Table 6: **Default memory module configuration.**

Hyperparameter	Setting
Local observation window k	16
Max tokens per frame N_f	100
Memory bank capacity N_b	100
Memory token count N_m	300
Frame consolidation	Sim-based merging
Spatial embedding	Fourier features
Temporal embedding	Sinusoidal embedding
ToMe layers	2
ToMe attention heads	16
Perceiver layers	2
Perceiver attention heads	16

Table 7: **MEMVLN training hyperparameters.**

Category	Setting
Hardware	$8 \times$ A100 80G GPUs
Precision	BF16
Training epochs	1
Per-device batch size	1
Gradient accumulation steps	1
Effective batch size	$1 \times 8 \times 1 = 8$
Optimizer	AdamW
Base learning rate	2×10^{-5}
Weight decay	0.05
Warmup steps	400
LoRA rank / alpha	128 / 256

B INFERENCE LATENCY AND COMPUTATIONAL COST

Evaluation Protocol. We measure system-level cost using fixed-input policy replay on one NVIDIA RTX 4090 with batch size 1 and the configured BF16/FlashAttention 2 inference setup. We compare the Base policy with the complete MEMVLN pipeline to measure the combined system overhead of StreamVGGT and episodic memory. The replay uses three frozen R2R episode traces comprising 102 fixed policy decisions. Each variant is evaluated in two fresh processes after the same 20-decision warm-up and state reset, yielding 204 timed observations per variant, including 168 observations in which the recent-context window contains 16 frames. Timing starts from RGB arrays in CPU memory and includes message construction, model preprocessing, host-to-device transfer, policy generation, decoding, and action parsing. It excludes sensor capture, Wi-Fi transfer, environment stepping, and the low-level robot controller. This fixed-input diagnostic characterizes policy cost and is not used to compute navigation metrics.

Table 8: **Fixed-replay inference latency and peak resource usage on one RTX 4090.** Latency statistics pool two fresh-process repeats of the same 102 fixed decisions; N denotes timed observations rather than independent episodes. The 16-frame recent-context subset contains 168 timed observations per variant. Policy rate is 1000/mean latency (ms). Resource rows report the maximum across valid repeats. Process GPU is the peak PID-scoped memory reported by `nvidia-smi`, and RSS is the peak host resident set size of the evaluation process.

Metric	Base	Full MEMVLN
Timed observations, N	204	204
Mean latency (ms)↓	503.2	691.0
P50 latency (ms)↓	535.2	704.3
P95 latency (ms)↓	550.0	779.8
Policy rate (Hz)↑	1.99	1.45
16-frame observations, N	168	168
16-frame mean latency (ms)↓	534.9	731.7
CUDA peak allocated (GiB)↓	10.725	17.104
CUDA peak reserved (GiB)↓	11.234	21.078
Process GPU peak (GiB)↓	11.69	21.54
Host RSS peak (GiB)↓	3.401	8.666

End-to-End Latency. As shown in Tab. 8, the complete pipeline increases mean policy latency from 503.2 to 691.0 ms (37.3%), corresponding to a sequential high-level policy rate of 1.45 Hz on the tested GPU. For the 16-frame recent-context subset, mean latency is 534.9 ms for Base and 731.7 ms for Full MEMVLN, a 36.8% increase.

Table 9: **Memory capacity ablations on R2R-CE and RxR-CE.**

(a) Memory bank capacity

N_b	R2R Val-Unseen				RxR Val-Unseen			
	NE↓	OSR↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑
0	5.87	57.4	50.9	47.2	5.72	54.9	**49.0**	64.5
100	**5.19**	**61.2**	**55.0**	**50.5**	**5.59**	**55.4**	48.4	**64.6**
200	6.24	58.7	50.2	45.7	6.15	54.3	48.0	61.5

(b) Memory-token count

N_m	R2R Val-Unseen				RxR Val-Unseen			
	NE↓	OSR↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑
0	5.87	57.4	50.9	47.2	5.72	54.9	**49.0**	64.5
300	**5.19**	**61.2**	**55.0**	**50.5**	**5.59**	**55.4**	48.4	**64.6**
600	6.75	53.9	45.1	40.8	6.64	50.4	44.0	59.3

Fixed Token Interface. For one R2R decision with a 16-frame recent-context window, Base receives 2,819 input tokens and Full MEMVLN receives 3,126. In both cases, 15 history frames are padded to 16 and grouped into eight temporal units; these units and the current image each contribute 300 visual tokens, for $(8 + 1) \times 300 = 2{,}700$ visual tokens. Full MEMVLN additionally contributes 300 fixed memory tokens and 7 associated prompt/special tokens, a fixed increase of 307 tokens under the evaluated prompt format. Other text tokens depend on the instruction. The bounded raw bank is not concatenated into the VLM context, so this memory-side input increment does not grow with route length.

Control Frequency. The measured mean latency in Tab. 8 corresponds to a sequential high-level policy rate of 1.45 Hz under the tested configuration. The real-robot client uses a separate low-level PID control thread to send commands through the Unitree SDK, nominally configured at approximately 30 Hz. The RTX 4090 benchmark measures high-level policy inference. We use *streaming* to denote causal online processing with bounded state.

Training Configuration. The main paper reports that the simulation model has approximately 1B trainable parameters and is trained for one epoch, taking approximately two days on 8×A100 80G GPUs. Additional training hyperparameters are summarized in Tab. 7.

C Additional Results

C.1 Memory Capacity

Table 9 reports the memory bank capacity and memory-token count ablations on both benchmarks. The default configuration uses $N_b = 100$, $N_m = 300$, and $N_f = 100$ patch tokens per stored frame. The zero-capacity setting retains the geometry encoder and trajectory-summary supervision.

C.2 Long-Horizon Analysis Protocol

We apply the same instruction-based grouping to all three model variants over 3,669 RxR-CE Val-Unseen episodes in Fig. 3. For instruction complexity, we count the total occurrences of commas and periods in each instruction as a punctuation-based proxy. The four bins contain 1–3, 4–6, 7–10, and more than 10 punctuation marks, comprising 239, 819, 1,113, and 1,498 episodes, respectively.

Instruction-type subsets are selected by case-insensitive keyword matching with word boundaries. The Back subset matches `turn around`, `back`, or `behind`, yielding 1,272 episodes. The Stop subset matches `stop`, `wait`, `end`, `hallway`, `hall`, or `corridor`, yielding 2,072 episodes. The Pass subset matches `pass`, `past`, `through`, or `cross`, yielding 1,410 episodes. An instruction enters a subset when it contains at least one corresponding keyword or phrase. These subsets are constructed independently and can overlap; All denotes the complete set of 3,669 episodes.

Table 10: **Real-world evaluation instructions.** We use the instruction IDs in subsequent results.

ID	Instruction
I1	Now you are facing a brown curtain. Turn right and walk pass an armchair towards the fire extinguisher. Turn left and walk toward the glass door in front of you to the right, that's your destination.
I2	Now you are facing the reception. On your left you should see some brown sofas. Walk past the sofas to the handwriting on the wall. Stop at the door on your right.
I3	Now you are in a corridor. Keep moving forward. Once you pass the bookshelf on the right side, turn right and go to the sink in the corner.
I4	Go straight towards the glass windows and past the sofas on your right. After you reach the windows, turn right. Keep moving forward and stop at the second door on your right.
I5	Now you are in a corridor. On your left, there are some white cabinets. Go past them and then turn left. You should see a red fire extinguisher on the ground. That box is your destination.
I6	Now you are in a lab. On your right there are some wooden stakes. Move forward and pass the chairs in front of you. You should see a robot sitting on a chair. Walk past the robot and stop at the bucket next to the lamp.

Table 11: **Per-instruction real-world results.** Each cell reports SR / CR (%) over five repeated trials for the corresponding instruction and method. Bold marks the best nonzero SR and lowest CR in each row.

ID	Base	+Mem	MemVLN	MemVLN-PT
I1	20 / 40	40 / 60	**80** / **20**	60 / 80
I2	0 / 60	0 / 60	**20** / 60	**20** / **40**
I3	40 / 60	**80** / 60	40 / **40**	**80** / 60
I4	0 / 100	0 / 100	0 / **60**	0 / **60**
I5	20 / **20**	0 / 40	20 / 40	**60** / 40
I6	20 / 60	60 / 80	**80** / 60	60 / **0**

C.3 Real-world Results

Evaluation Protocol. We provide a detailed breakdown of the real-world G1 evaluation summarized in Sec. 4.4. All 30 instruction-conditioned episodes were collected in the same physical scene and split at the episode level: 24 demonstrations are used for post-training and six distinct episodes are held out for evaluation. We evaluate Base, +Mem, MemVLN, and MemVLN-PT, repeating each held-out instruction five times for 30 trials per method. In addition to success rate (SR), we report collision rate (CR), the fraction of trials involving physical contact, contact-induced blockage, or contact-induced termination; CR measures contact safety independently of task completion.

Deployment Setup. During real-world deployment, policy inference runs on a host workstation equipped with a single NVIDIA RTX 4090 GPU. RGB observations captured by the robot-mounted Intel RealSense D455 are transmitted to the host over Wi-Fi; after the host completes model inference, the predicted action is sent back to the G1 robot for execution.

Overall Performance. As shown in Fig. 4, Base succeeds in 5 of 30 runs, corresponding to 16.7% SR. Adding the memory module increases success to 9 of 30 runs (30.0% SR), and the full MemVLN model further improves success to 12 of 30 runs (40.0% SR). After real-world post-training with 24 demonstrations, MemVLN-PT succeeds in 14 of 30 runs (46.7% SR), giving two additional successful runs over MemVLN. CR remains unchanged at 46.7% for MemVLN and MemVLN-PT.

Instruction-level Analysis. As shown in Tab. 11, MemVLN-PT improves SR over MemVLN on I3 and I5, leaves it unchanged on I2 and I4, and decreases it on I1 and I6. Post-training lowers CR on I2 and I6 and increases it on I1 and I3. All four methods obtain 0% SR on I4, which requires selecting the second door on the right after passing an intermediate door.

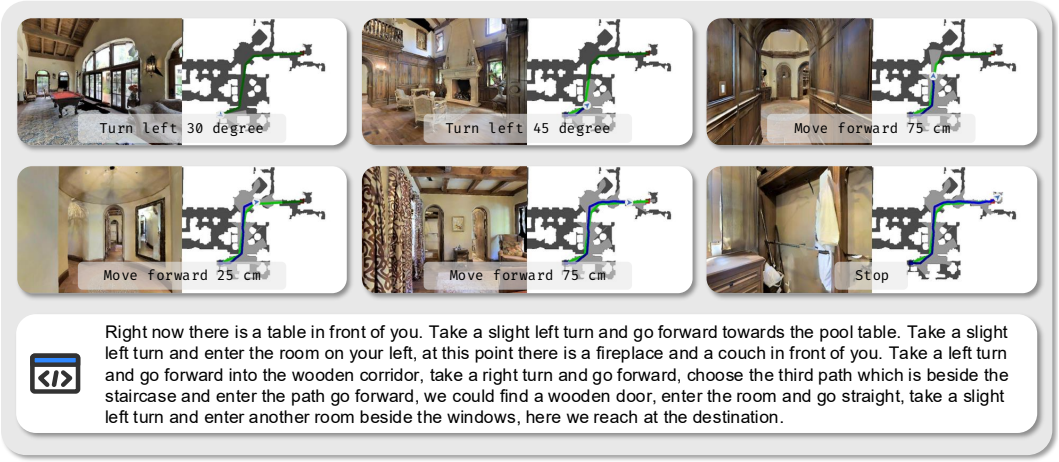
Remaining Challenges. Collision remains a challenge: MemVLN and MemVLN-PT both have a CR of 46.7%. Post-training improves aggregate SR but leaves aggregate CR unchanged. Future work will explore contact-aware control and richer supervision for fine-grained spatial grounding.

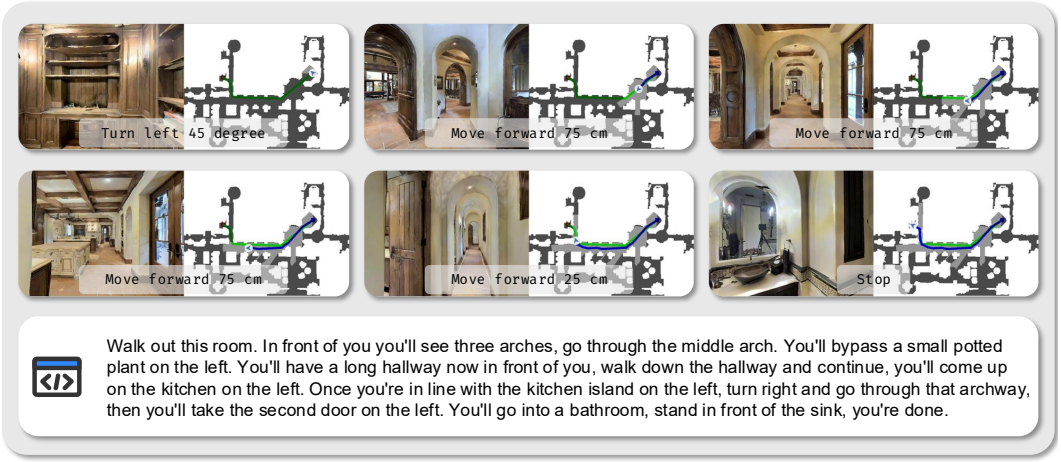
C.4 More Qualitative Results

Simulation Results. Figure 6 and 7 show R2R-CE and RxR-CE navigation rollouts, including selected observations and actions along multi-step routes.



Figure 6: **Qualitative results of MemVLN on R2R-CE.**





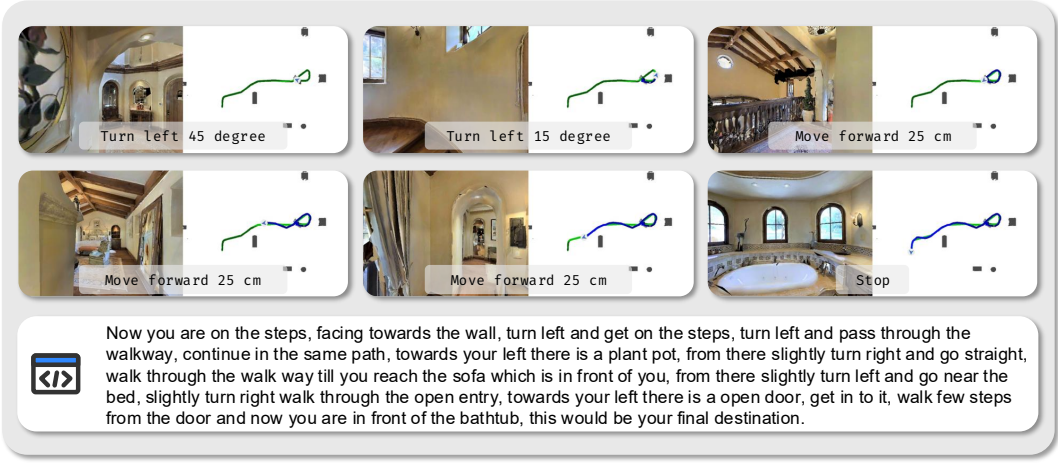


Figure 7: **Qualitative results of MemVLN on RxR-CE.**

Real-world Results. Figure 8 shows two additional G1 navigation rollouts with their instructions, selected observations, and actions.

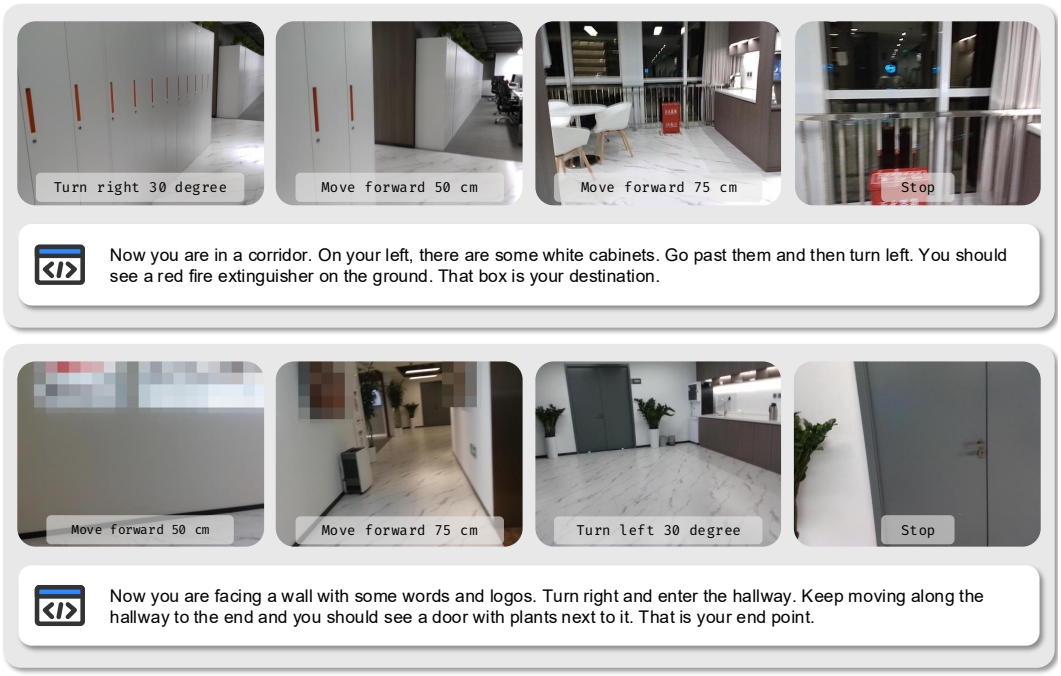


Figure 8: **Qualitative results of MEMVLN in the real world.** Selected observations and actions from two G1 navigation rollouts, together with their navigation instructions.